Results Supplement

Predicting NMR Chemical Shifts from Sequence

This supplement contains the complete recorded development controls and the residue-level tables underlying the reported chemical-shift comparisons. All model errors are calculated from exact assignments and reported in parts per million (ppm). H, HA, CA, CB, C and N are reported separately; HA contains HA, HA2 and HA3. Side-chain H, C and N are also separate in the broad-transfer table. No pooled H/C/N score or selector loss is reported.

For calculated shielding, T_0 is the isotropic component. The five T_2 values use an orthonormal symmetric-traceless basis; their RMS and mean norms are therefore the corresponding tensor Frobenius-error norms. For broad RefDB transfer, the reference is the arithmetic residue–atom mean fitted only to the 173 training proteins, with the recorded element and global fallbacks. Residual r is Pearson correlation after subtracting that reference from both prediction and observation.

Contents

1 Calculated-Shielding Results

ShieldingTfnModel condition	Atom	Small AlphaFold Shielding Set: fitting (521 proteins)			Small AlphaFold Shielding Set: selector (25 proteins)			Small AlphaFold Shielding Set: protected (79 proteins)		
		T_0 RMSE	T_2 rec.	cosine	T_0 RMSE	T_2 rec.	cosine	T_0 RMSE	T_2 rec.	cosine
Typed Graph	Cα	0.484	91.9%	0.9968	0.648	90.8%	0.9960	0.626	90.9%	0.9959
	Cβ	0.444	90.6%	0.9955	0.654	88.8%	0.9938	0.655	89.2%	0.9940
	C$'$	0.367	96.1%	0.9994	0.518	96.1%	0.9994	0.550	95.9%	0.9993
	O	2.364	96.2%	0.9995	2.932	95.9%	0.9994	3.186	95.9%	0.9994
	N	0.937	96.3%	0.9991	1.384	95.7%	0.9988	1.304	95.8%	0.9989
	Hα	0.076	90.1%	0.9942	0.093	88.0%	0.9916	0.087	88.7%	0.9925
	H^N	0.135	92.9%	0.9975	0.152	92.0%	0.9967	0.153	92.0%	0.9970
AIMNet2	Cα	0.426	92.5%	0.9973	0.614	91.3%	0.9966	0.613	91.2%	0.9962
	Cβ	0.383	91.4%	0.9962	0.626	89.2%	0.9941	0.677	89.7%	0.9945
	C$'$	0.330	96.4%	0.9995	0.539	96.1%	0.9994	0.563	96.0%	0.9994
	O	2.158	96.3%	0.9995	3.156	95.9%	0.9994	3.097	95.9%	0.9994
	N	0.842	96.4%	0.9992	1.320	95.6%	0.9985	1.252	95.8%	0.9989
	Hα	0.071	90.7%	0.9950	0.088	88.7%	0.9928	0.090	89.1%	0.9931
	H^N	0.120	93.5%	0.9979	0.149	92.5%	0.9972	0.146	92.5%	0.9973
MOPAC	Cα	0.406	92.6%	0.9973	0.574	91.6%	0.9967	0.594	91.2%	0.9960
	Cβ	0.404	91.5%	0.9963	0.621	89.5%	0.9946	0.667	89.4%	0.9944
	C$'$	0.323	96.4%	0.9995	0.490	96.3%	0.9995	0.516	95.8%	0.9994
	O	2.234	96.5%	0.9996	3.036	96.4%	0.9995	3.266	96.2%	0.9995
	N	0.884	96.5%	0.9993	1.193	96.0%	0.9987	1.346	95.7%	0.9989
	Hα	0.065	91.6%	0.9958	0.083	89.9%	0.9939	0.083	89.4%	0.9937
	H^N	0.114	93.8%	0.9981	0.138	93.0%	0.9975	0.142	92.8%	0.9974
AIMNet2+MOPAC	Cα	0.378	93.3%	0.9978	0.563	92.2%	0.9972	0.619	91.4%	0.9964
	Cβ	0.321	92.2%	0.9969	0.564	89.9%	0.9950	0.625	89.3%	0.9944
	C$'$	0.301	96.7%	0.9996	0.498	96.3%	0.9995	0.511	96.1%	0.9994
	O	1.879	96.8%	0.9996	2.771	96.5%	0.9995	2.990	96.3%	0.9995
	N	0.804	96.7%	0.9994	1.269	96.1%	0.9989	1.293	96.0%	0.9991
	Hα	0.060	91.8%	0.9961	0.078	89.6%	0.9938	0.081	89.6%	0.9939
	H^N	0.105	94.2%	0.9983	0.144	93.1%	0.9976	0.132	93.1%	0.9977

T_0 RMSE is in shielding ppm. T_2 recovery is $1 - \mathrm{RMSE}_{\mathrm{model}}/\mathrm{RMSE}_{\mathrm{zero\ tensor}}$; cosine is the mean normalized dot product.

Table S1: Backbone shielding performance across the three **Small AlphaFold Shielding Set** partitions.

ShieldingTfnModel condition	Atom	TrajectoryTest1P9JSet (751 frames)			TrajectoryTest1L2YSet (1,501 frames)			TrajectoryTest5AWLSet (5,001 frames)		
		T_0 RMSE	T_2 rec.	cosine	T_0 RMSE	T_2 rec.	cosine	T_0 RMSE	T_2 rec.	cosine
Typed Graph	Cα	3.819	69.2%	0.9517	3.813	69.8%	0.9484	3.983	67.4%	0.9445
	Cβ	3.525	63.7%	0.9252	3.007	68.6%	0.9499	3.276	64.5%	0.9355
	C$'$	4.289	89.1%	0.9953	4.108	89.8%	0.9958	4.214	89.4%	0.9954
	N	6.637	85.9%	0.9879	6.366	86.4%	0.9792	6.249	85.5%	0.9432
	Hα	0.310	78.0%	0.9717	0.280	79.9%	0.9770	0.289	78.7%	0.9779
	H^N	0.425	85.3%	0.9894	0.387	86.7%	0.9924	0.428	84.3%	0.9873
AIMNet2	Cα	3.787	71.1%	0.9578	3.698	70.9%	0.9545	3.839	70.6%	0.9575
	Cβ	3.739	65.0%	0.9304	2.882	68.5%	0.9510	3.154	66.2%	0.9422
	C$'$	4.304	89.4%	0.9960	4.249	89.7%	0.9962	4.365	89.8%	0.9961
	N	6.552	86.0%	0.9889	6.655	86.9%	0.9855	9.600	85.7%	0.9553
	Hα	0.279	78.6%	0.9726	0.265	79.5%	0.9754	0.270	79.3%	0.9782
	H^N	0.427	85.2%	0.9890	0.397	86.0%	0.9915	0.403	84.8%	0.9881
MOPAC	Cα	3.901	55.3%	0.9029	3.525	71.8%	0.9563	3.765	70.2%	0.9557
	Cβ	4.244	51.9%	0.8931	2.947	69.7%	0.9546	3.170	66.6%	0.9404
	C$'$	4.964	78.5%	0.9845	4.518	89.0%	0.9956	4.529	88.1%	0.9946
	N	8.827	77.1%	0.9757	6.970	86.3%	0.9830	6.659	85.4%	0.9487
	Hα	0.427	55.1%	0.8904	0.276	80.0%	0.9784	0.322	78.3%	0.9749
	H^N	0.785	60.9%	0.9558	0.458	85.4%	0.9916	0.493	84.3%	0.9869
AIMNet2+MOPAC	Cα	3.877	58.2%	0.9154	3.813	71.7%	0.9574	3.885	70.6%	0.9569
	Cβ	4.089	51.1%	0.8883	2.983	70.3%	0.9575	3.321	68.6%	0.9470
	C$'$	4.783	81.2%	0.9863	4.399	90.1%	0.9964	4.393	89.9%	0.9962
	N	7.562	79.7%	0.9819	6.115	86.6%	0.9812	6.718	85.7%	0.9569
	Hα	0.478	53.2%	0.8774	0.269	80.5%	0.9788	0.282	79.4%	0.9777
	H^N	0.703	67.9%	0.9424	0.414	87.2%	0.9928	0.464	86.3%	0.9907

T_0 RMSE is in shielding ppm. T_2 recovery is $1 - \mathrm{RMSE}_{\mathrm{model}}/\mathrm{RMSE}_{\mathrm{zero\ tensor}}$; cosine is the mean normalized dot product.

Table S2: Backbone shielding performance on the three trajectory test sets.

2 Chemical-Shift Development Controls

Input representation	Model form	H	HA	CA	CB	C	N
Typed Graph	Static Models	0.368	0.188	0.755	0.898	1.134	1.983
Typed Graph	Single-Frame MD Models	0.446	0.228	0.968	1.048	1.251	2.244
MOPAC	Static Models	0.358	0.184	0.746	0.873	1.132	1.943
MOPAC	Single-Frame MD Models	0.440	0.230	0.941	1.033	1.275	2.475
AIMNet2	Static Models	0.399	0.211	0.879	0.977	1.235	2.394
AIMNet2	Single-Frame MD Models	0.440	0.230	0.978	1.048	1.215	2.230
AIMNet2+MOPAC	Static Models	0.401	0.205	0.876	1.002	1.215	2.521
AIMNet2+MOPAC	Single-Frame MD Models	0.451	0.225	0.995	1.067	1.198	2.352

Table S3: Exact-assignment backbone RMSE (ppm) on the 22 selector proteins of the **RefDB Curated GROMACS Set**.

Model condition	H	HA	CA	CB	C	N
Static Models	0.368	0.188	0.755	0.898	1.134	1.983
Static Models without OpenFold3	0.381	0.194	0.870	0.920	1.165	2.012
Static Models without calculated-shielding information	0.434	0.237	0.840	0.957	1.226	2.640
Single-Frame MD Models	0.446	0.228	0.968	1.048	1.251	2.244
Single-Frame MD Models without calculated-shielding information	0.490	0.285	1.009	1.076	1.313	2.837

Table S4: Exact-assignment backbone RMSE (ppm) on the 22 selector proteins of the **RefDB Curated GROMACS Set**. The calculated-shielding-information control jointly replaced the shielding-trained chemical encoder with a randomly initialised encoder and removed predicted T_0, T_2 magnitude and T_2 shape, ring current and the five McConnell inputs.

100-frame model condition	H	HA	CA	CB	C	N
100-Frame MD Model	0.4132	0.2091	0.8651	0.9861	1.2004	2.1003
without frozen-parent response	0.4061	0.2078	0.8600	0.9746	1.1775	2.1003
without scalar structure and electronic quantities	0.4122	0.2079	0.8651	0.9861	1.2004	2.0995
without vector and rank-2 invariants	0.4121	0.2087	0.8651	0.9861	1.2004	2.0979
without shielding and physical values	0.4132	0.2095	0.8651	0.9861	1.2004	2.1001
without torsion and rotamer quantities	0.4140	0.2092	0.8651	0.9861	1.2004	2.1006
without aligned atomic motion	0.4132	0.2088	0.8651	0.9861	1.2004	2.1001
without support and DSSP state	0.4135	0.2090	0.8651	0.9861	1.2004	2.1004

Table S5: Exact-assignment backbone RMSE (ppm) on the 22 selector proteins of the **RefDB Curated GROMACS Set**. Each row refitted the seed-0 **100-Frame MD Model** after suppressing the named input family.

3 Complete Calculated-Shielding Input Matrix

Input condition	Primary reference	Atoms	Proteins	T_0 RMSE	T_0 MAE	T_2 RMS norm	T_2 mean norm
Structure only	—	6,524	25	1.130	0.682	4.871	3.709
AIMNet embedding	Structure only	6,524	25	0.987	0.587	4.432	3.356
AIMNet embedding + electrostatics	AIMNet embedding	6,524	25	1.001	0.585	4.393	3.326
AIMNet embedding + charge response	AIMNet embedding	6,524	25	0.984	0.583	4.339	3.297
AIMNet embedding + energy/dispersion	AIMNet embedding	6,524	25	1.002	0.596	4.367	3.310
All AIMNet quantities	AIMNet embedding	6,524	25	1.003	0.587	4.354	3.287
MOPAC inputs without AIMNet	Structure only	6,524	25	1.014	0.611	4.255	3.258
AIMNet embedding + MOPAC atomic state	AIMNet embedding	6,524	25	0.992	0.589	4.428	3.350
AIMNet embedding + MOPAC density/bonding	AIMNet embedding	6,524	25	0.981	0.568	4.142	3.164
AIMNet embedding + MOPAC electrostatics	AIMNet embedding	6,524	25	0.973	0.576	4.429	3.354
AIMNet embedding + peptide-carbonyl response	AIMNet embedding	6,524	25	1.032	0.591	4.408	3.349
AIMNet embedding + all selected MOPAC inputs	AIMNet embedding	6,524	25	0.927	0.560	4.178	3.165
AIMNet embedding + ring current	AIMNet embedding	6,524	25	1.007	0.606	4.541	3.450
AIMNet embedding + hydrogen-bond geometry	AIMNet embedding	6,524	25	1.010	0.607	4.434	3.386
AIMNet embedding + solvent exposure	AIMNet embedding	6,524	25	1.009	0.598	4.410	3.319
AIMNet embedding + fixed McConnell terms	AIMNet embedding	6,524	25	0.995	0.577	4.434	3.366
AIMNet embedding + APBS quantities	AIMNet embedding	6,524	25	0.986	0.596	4.401	3.333
AIMNet embedding + environment calculations	AIMNet embedding	6,524	25	1.044	0.591	4.657	3.554
AIMNet embedding + MOPAC + environment	AIMNet embedding + all selected MOPAC inputs	6,524	25	0.923	0.543	4.311	3.268
All AIMNet + MOPAC + environment quantities	AIMNet embedding + MOPAC + environment	6,524	25	0.919	0.546	4.232	3.197

Table S6: Calculated-shielding results for the complete 20-condition selector matrix. All errors are in ppm; T_2 columns are error-vector norms in the orthonormal five-component basis.

4 Protected Chemical-Shift Results

Model route	H	HA	CA	CB	C	N
Static Models	0.369	0.196	0.811	0.884	0.889	2.375
Single-Frame MD Models	0.442	0.230	0.991	1.058	1.030	2.701
100-Frame MD Model, seed 0	0.401	0.206	0.921	0.955	0.961	2.589

Table S7: Exact-assignment backbone RMSE (ppm) on the 45 protected proteins of the RefDB Curated GROMACS Set.

Model form	Encoder start	H	HA	CA	CB	C	N
Static Models	Shielding-pretrained	0.381	0.195	0.802	0.907	0.905	2.464
Static Models	Trained from scratch	0.424	0.213	0.892	0.988	1.026	2.815
Single-Frame MD Models	Shielding-pretrained	0.439	0.230	0.978	1.049	1.016	2.754
Single-Frame MD Models	Trained from scratch	0.467	0.249	1.093	1.146	1.142	3.193

Table S8: Protected backbone RMSE (ppm) for the matched shielding-encoder initialisation control. Both arms retained the same calculated physical inputs and trained the encoder normally.

4.1 Complete residue–backbone tables

Residue	Atom	Assignments	Proteins	Static	Single frame	Uniform 100-frame	Learned 100-frame mean	Learned SD
ALA	H	184	45	0.357	0.426	0.353	0.378	0.004
ALA	HA	182	43	0.167	0.189	0.193	0.180	0.002
ALA	CA	174	42	0.600	0.689	0.667	0.670	0.002
ALA	CB	164	42	0.736	0.897	0.802	0.807	0.009
ALA	C	134	35	0.912	1.072	0.955	0.976	0.021
ALA	N	171	44	1.923	2.433	2.296	2.296	0.001
ARG	H	191	42	0.332	0.360	0.330	0.342	0.001
ARG	HA	187	40	0.181	0.227	0.197	0.207	0.000
ARG	CA	184	40	0.851	1.043	0.943	0.941	0.002
ARG	CB	172	40	0.790	0.938	0.761	0.761	0.005
ARG	C	140	33	0.819	1.081	1.011	1.014	0.009
ARG	N	169	41	2.250	2.329	2.445	2.445	0.001
ASN	H	119	42	0.378	0.466	0.439	0.441	0.001
ASN	HA	120	41	0.182	0.218	0.204	0.208	0.004
ASN	CA	114	39	0.861	0.957	0.902	0.915	0.012
ASN	CB	109	39	0.773	0.891	0.877	0.872	0.004
ASN	C	72	30	0.828	1.007	0.895	0.928	0.030
ASN	N	111	39	2.513	2.692	2.606	2.604	0.003
ASP	H	182	43	0.395	0.422	0.395	0.404	0.003
ASP	HA	177	41	0.146	0.186	0.157	0.163	0.001
ASP	CA	171	40	0.918	1.083	1.027	1.020	0.008
ASP	CB	162	40	1.157	1.260	1.243	1.242	0.004
ASP	C	123	33	0.741	0.904	0.853	0.862	0.008
ASP	N	166	42	2.190	2.460	2.355	2.355	0.001
CYS	H	27	10	0.471	0.695	0.575	0.633	0.010
CYS	HA	27	10	0.252	0.294	0.279	0.255	0.016
CYS	CA	27	10	1.645	1.796	1.774	1.776	0.015
CYS	CB	26	10	2.779	2.811	2.736	2.711	0.021
CYS	C	15	6	1.353	1.407	1.399	1.383	0.027
CYS	N	24	9	2.914	3.424	3.431	3.426	0.008
GLN	H	181	42	0.336	0.423	0.369	0.376	0.002
GLN	HA	172	40	0.236	0.257	0.227	0.236	0.004
GLN	CA	170	39	0.707	0.857	0.818	0.816	0.002
GLN	CB	166	39	0.759	0.946	0.858	0.854	0.004
GLN	C	131	32	0.842	1.020	0.951	0.934	0.016
GLN	N	165	41	2.110	2.537	2.469	2.470	0.002
GLU	H	232	45	0.420	0.481	0.413	0.432	0.004
GLU	HA	222	43	0.189	0.226	0.194	0.194	0.004
GLU	CA	207	42	0.734	0.863	0.763	0.768	0.004
GLU	CB	196	41	0.759	1.010	0.908	0.922	0.013
GLU	C	165	35	0.882	0.987	0.967	0.965	0.003
GLU	N	215	44	2.069	2.502	2.172	2.173	0.002
GLY	H	260	45	0.380	0.447	0.368	0.386	0.002
GLY	HA	150	30	0.211	0.193	0.196	0.200	0.002
GLY	CA	238	40	0.599	0.729	0.676	0.681	0.008
GLY	C	177	32	1.066	1.089	1.070	1.057	0.035
GLY	N	225	41	3.244	3.305	3.193	3.190	0.005

continued

Residue	Atom	Assignments	Proteins	Static	Single frame	Uniform 100-frame	Learned 100-frame mean	Learned SD
HIS	H	57	29	0.443	0.371	0.388	0.381	0.011
HIS	HA	59	27	0.301	0.272	0.239	0.261	0.005
HIS	CA	58	27	1.057	1.142	1.082	1.077	0.005
HIS	CB	55	26	0.844	0.940	0.888	0.894	0.005
HIS	C	39	20	0.791	0.788	0.829	0.818	0.013
HIS	N	53	28	3.161	3.607	3.517	3.518	0.002
ILE	H	179	44	0.363	0.525	0.474	0.486	0.003
ILE	HA	165	42	0.195	0.263	0.258	0.248	0.007
ILE	CA	157	41	0.823	0.990	1.049	0.998	0.045
ILE	CB	153	41	0.958	1.181	1.150	1.133	0.015
ILE	C	116	34	0.862	0.994	0.977	0.976	0.004
ILE	N	163	43	2.429	2.670	2.562	2.562	0.001
LEU	H	327	45	0.380	0.458	0.417	0.420	0.001
LEU	HA	308	43	0.172	0.246	0.220	0.210	0.004
LEU	CA	296	42	0.690	0.855	0.732	0.746	0.012
LEU	CB	285	42	0.873	1.199	1.052	1.061	0.011
LEU	C	225	35	0.878	0.992	0.918	0.926	0.007
LEU	N	302	44	2.142	2.453	2.341	2.340	0.001
LYS	H	254	45	0.308	0.367	0.329	0.341	0.003
LYS	HA	237	43	0.183	0.240	0.203	0.207	0.003
LYS	CA	216	42	0.863	1.069	0.947	0.946	0.001
LYS	CB	200	42	0.697	0.949	0.773	0.787	0.014
LYS	C	163	35	0.765	0.949	0.860	0.840	0.017
LYS	N	221	44	2.149	2.264	2.202	2.201	0.002
MET	H	46	26	0.349	0.356	0.338	0.334	0.007
MET	HA	54	30	0.237	0.273	0.236	0.238	0.009
MET	CA	50	28	0.822	1.251	0.998	1.003	0.017
MET	CB	44	28	1.470	1.611	1.389	1.398	0.012
MET	C	35	22	1.127	1.168	1.091	1.092	0.014
MET	N	41	24	1.891	2.093	1.851	1.853	0.003
PHE	H	105	43	0.353	0.472	0.396	0.394	0.009
PHE	HA	103	41	0.220	0.288	0.272	0.263	0.001
PHE	CA	97	40	0.919	1.171	1.187	1.184	0.008
PHE	CB	93	40	0.889	1.171	1.044	1.045	0.001
PHE	C	81	34	0.944	1.125	1.065	1.066	0.002
PHE	N	99	42	1.827	2.036	1.839	1.836	0.005
PRO	HA	158	43	0.186	0.225	0.213	0.200	0.006
PRO	CA	139	42	0.598	0.671	0.584	0.587	0.017
PRO	CB	135	42	0.757	0.775	0.733	0.734	0.004
PRO	C	98	34	0.993	1.050	0.990	1.003	0.013
SER	H	198	45	0.384	0.446	0.435	0.425	0.002
SER	HA	223	43	0.175	0.186	0.164	0.167	0.003
SER	CA	190	41	0.914	1.163	1.066	1.074	0.007
SER	CB	182	41	0.680	0.711	0.664	0.661	0.003
SER	C	143	33	0.911	1.067	0.949	0.966	0.016
SER	N	157	43	2.248	2.658	2.513	2.513	0.000
THR	H	172	43	0.356	0.439	0.381	0.394	0.002
THR	HA	166	41	0.189	0.244	0.210	0.203	0.002
THR	CA	166	40	0.675	0.971	1.006	0.976	0.028
THR	CB	160	40	0.767	0.889	0.762	0.774	0.015
THR	C	116	32	0.799	0.958	0.906	0.910	0.011

continued

Residue	Atom	Assignments	Proteins	Static	Single frame	Uniform 100-frame	Learned 100-frame mean	Learned SD
THR	N	157	42	3.367	4.005	3.997	3.996	0.001
TRP	H	50	28	0.464	0.489	0.415	0.447	0.011
TRP	HA	50	28	0.199	0.247	0.196	0.203	0.003
TRP	CA	48	28	1.179	1.490	1.296	1.337	0.039
TRP	CB	48	28	0.912	1.192	1.105	1.100	0.005
TRP	C	37	25	1.210	1.395	1.231	1.280	0.042
TRP	N	47	28	2.142	2.043	2.367	2.364	0.005
TYR	H	92	44	0.362	0.465	0.396	0.412	0.001
TYR	HA	91	41	0.214	0.203	0.195	0.180	0.002
TYR	CA	91	41	0.945	1.146	1.175	1.149	0.024
TYR	CB	88	41	0.909	1.196	0.907	0.934	0.025
TYR	C	70	33	0.762	0.961	0.845	0.853	0.009
TYR	N	87	42	1.755	2.410	2.044	2.040	0.007
VAL	H	182	44	0.324	0.414	0.347	0.364	0.003
VAL	HA	174	42	0.237	0.216	0.190	0.184	0.001
VAL	CA	169	41	0.932	1.153	0.953	0.960	0.009
VAL	CB	160	41	0.837	0.942	0.829	0.815	0.012
VAL	C	127	34	0.849	1.055	0.952	0.962	0.009
VAL	N	165	42	2.204	2.880	2.656	2.651	0.008

Table S9: Protected residue–backbone RMSE (ppm) for the Static, Single-Frame MD, uniform 100-frame and learned 100-Frame MD routes. Learned values are the mean and sample standard deviation across three seeds.

Residue	Atom	Assignments	Proteins	Static	Single frame	Uniform 100-frame	Learned 100-frame mean	Learned SD
ALA	H	184	45	0.272	0.339	0.273	0.293	0.004
ALA	HA	182	43	0.125	0.138	0.142	0.132	0.003
ALA	CA	174	42	0.464	0.530	0.515	0.515	0.001
ALA	CB	164	42	0.516	0.653	0.586	0.588	0.010
ALA	C	134	35	0.709	0.799	0.731	0.742	0.014
ALA	N	171	44	1.462	1.851	1.720	1.719	0.001
ARG	H	191	42	0.248	0.277	0.252	0.267	0.005
ARG	HA	187	40	0.131	0.169	0.141	0.147	0.002
ARG	CA	184	40	0.632	0.786	0.688	0.684	0.004
ARG	CB	172	40	0.577	0.701	0.594	0.588	0.006
ARG	C	140	33	0.653	0.806	0.766	0.763	0.006
ARG	N	169	41	1.658	1.817	1.906	1.906	0.001
ASN	H	119	42	0.244	0.312	0.274	0.302	0.005
ASN	HA	120	41	0.136	0.156	0.148	0.150	0.002
ASN	CA	114	39	0.690	0.767	0.689	0.705	0.015
ASN	CB	109	39	0.577	0.682	0.686	0.678	0.007
ASN	C	72	30	0.651	0.788	0.691	0.723	0.028
ASN	N	111	39	1.937	2.130	1.997	1.996	0.002
ASP	H	182	43	0.273	0.316	0.290	0.300	0.002
ASP	HA	177	41	0.114	0.138	0.126	0.127	0.002
ASP	CA	171	40	0.674	0.819	0.777	0.769	0.008
ASP	CB	162	40	0.692	0.789	0.757	0.757	0.006
ASP	C	123	33	0.565	0.708	0.627	0.638	0.012
ASP	N	166	42	1.745	1.959	1.876	1.877	0.002
CYS	H	27	10	0.373	0.525	0.446	0.472	0.009
CYS	HA	27	10	0.215	0.228	0.209	0.199	0.012
CYS	CA	27	10	1.192	1.464	1.396	1.413	0.016
CYS	CB	26	10	1.941	2.079	2.003	1.996	0.010
CYS	C	15	6	1.116	1.137	1.078	1.074	0.039
CYS	N	24	9	2.022	2.466	2.183	2.184	0.002
GLN	H	181	42	0.249	0.325	0.269	0.284	0.003
GLN	HA	172	40	0.165	0.183	0.172	0.173	0.003
GLN	CA	170	39	0.577	0.692	0.648	0.646	0.005
GLN	CB	166	39	0.604	0.755	0.685	0.684	0.001
GLN	C	131	32	0.668	0.834	0.771	0.752	0.017
GLN	N	165	41	1.675	1.994	1.943	1.945	0.004
GLU	H	232	45	0.298	0.357	0.301	0.313	0.003
GLU	HA	222	43	0.135	0.168	0.142	0.144	0.003
GLU	CA	207	42	0.605	0.668	0.592	0.598	0.005
GLU	CB	196	41	0.577	0.744	0.679	0.689	0.009
GLU	C	165	35	0.693	0.805	0.768	0.754	0.012
GLU	N	215	44	1.524	1.865	1.728	1.728	0.000
GLY	H	260	45	0.269	0.307	0.254	0.273	0.001
GLY	HA	150	30	0.128	0.136	0.146	0.149	0.004
GLY	CA	238	40	0.423	0.523	0.471	0.479	0.008
GLY	C	177	32	0.646	0.749	0.698	0.709	0.014
GLY	N	225	41	1.876	2.037	1.963	1.959	0.007
HIS	H	57	29	0.319	0.281	0.299	0.293	0.011
HIS	HA	59	27	0.188	0.206	0.177	0.193	0.002
HIS	CA	58	27	0.826	0.900	0.863	0.845	0.017

continued

Residue	Atom	Assignments	Proteins	Static	Single frame	Uniform 100-frame	Learned 100-frame mean	Learned SD
HIS	CB	55	26	0.718	0.796	0.715	0.734	0.016
HIS	C	39	20	0.651	0.642	0.611	0.617	0.013
HIS	N	53	28	2.367	2.892	2.674	2.677	0.004
ILE	H	179	44	0.279	0.393	0.356	0.370	0.003
ILE	HA	165	42	0.148	0.207	0.197	0.187	0.004
ILE	CA	157	41	0.683	0.785	0.807	0.775	0.028
ILE	CB	153	41	0.749	0.932	0.884	0.865	0.016
ILE	C	116	34	0.657	0.790	0.781	0.775	0.005
ILE	N	163	43	1.886	2.059	2.011	2.011	0.000
LEU	H	327	45	0.286	0.360	0.320	0.321	0.003
LEU	HA	308	43	0.130	0.181	0.156	0.151	0.002
LEU	CA	296	42	0.497	0.664	0.549	0.563	0.013
LEU	CB	285	42	0.660	0.949	0.805	0.818	0.013
LEU	C	225	35	0.684	0.788	0.709	0.725	0.014
LEU	N	302	44	1.589	1.866	1.796	1.795	0.001
LYS	H	254	45	0.227	0.290	0.258	0.269	0.001
LYS	HA	237	43	0.138	0.173	0.147	0.147	0.002
LYS	CA	216	42	0.602	0.779	0.689	0.684	0.005
LYS	CB	200	42	0.493	0.702	0.595	0.601	0.007
LYS	C	163	35	0.618	0.730	0.683	0.663	0.018
LYS	N	221	44	1.674	1.768	1.721	1.719	0.003
MET	H	46	26	0.282	0.289	0.271	0.270	0.006
MET	HA	54	30	0.178	0.202	0.186	0.191	0.004
MET	CA	50	28	0.649	1.019	0.823	0.816	0.029
MET	CB	44	28	0.868	1.213	0.984	1.006	0.022
MET	C	35	22	0.855	0.934	0.880	0.880	0.016
MET	N	41	24	1.485	1.670	1.453	1.460	0.012
PHE	H	105	43	0.263	0.368	0.312	0.308	0.010
PHE	HA	103	41	0.178	0.230	0.218	0.209	0.002
PHE	CA	97	40	0.722	0.937	0.901	0.914	0.012
PHE	CB	93	40	0.674	0.878	0.815	0.808	0.007
PHE	C	81	34	0.792	0.919	0.884	0.882	0.002
PHE	N	99	42	1.441	1.598	1.355	1.352	0.005
PRO	HA	158	43	0.112	0.154	0.138	0.133	0.002
PRO	CA	139	42	0.471	0.529	0.474	0.465	0.012
PRO	CB	135	42	0.550	0.548	0.530	0.526	0.004
PRO	C	98	34	0.793	0.796	0.790	0.786	0.004
SER	H	198	45	0.274	0.330	0.302	0.298	0.002
SER	HA	223	43	0.126	0.130	0.114	0.117	0.004
SER	CA	190	41	0.697	0.850	0.764	0.768	0.008
SER	CB	182	41	0.500	0.530	0.494	0.491	0.003
SER	C	143	33	0.690	0.806	0.702	0.720	0.016
SER	N	157	43	1.709	1.823	1.763	1.763	0.001
THR	H	172	43	0.269	0.335	0.285	0.296	0.004
THR	HA	166	41	0.148	0.189	0.163	0.158	0.002
THR	CA	166	40	0.504	0.744	0.756	0.730	0.024
THR	CB	160	40	0.514	0.661	0.549	0.555	0.009
THR	C	116	32	0.574	0.698	0.642	0.656	0.018
THR	N	157	42	2.276	2.749	2.839	2.838	0.003
TRP	H	50	28	0.360	0.384	0.320	0.363	0.014
TRP	HA	50	28	0.154	0.190	0.154	0.153	0.003

continued

Residue	Atom	Assignments	Proteins	Static	Single frame	Uniform 100-frame	Learned 100-frame mean	Learned SD
TRP	CA	48	28	0.845	1.147	1.016	1.055	0.037
TRP	CB	48	28	0.724	0.952	0.906	0.896	0.008
TRP	C	37	25	1.030	1.130	1.090	1.118	0.025
TRP	N	47	28	1.644	1.637	1.901	1.898	0.006
TYR	H	92	44	0.264	0.365	0.305	0.321	0.005
TYR	HA	91	41	0.147	0.155	0.150	0.137	0.002
TYR	CA	91	41	0.690	0.887	0.874	0.859	0.012
TYR	CB	88	41	0.696	0.914	0.693	0.721	0.026
TYR	C	70	33	0.612	0.774	0.653	0.657	0.004
TYR	N	87	42	1.345	1.824	1.583	1.580	0.005
VAL	H	182	44	0.240	0.318	0.259	0.284	0.002
VAL	HA	174	42	0.183	0.173	0.153	0.146	0.001
VAL	CA	169	41	0.717	0.884	0.728	0.724	0.006
VAL	CB	160	41	0.688	0.749	0.673	0.662	0.011
VAL	C	127	34	0.627	0.841	0.720	0.735	0.014
VAL	N	165	42	1.718	2.200	2.087	2.083	0.007

Table S10: Protected residue–backbone MAE (ppm) for the Static, Single-Frame MD, uniform 100-frame and learned 100-Frame MD routes. Learned values are the mean and sample standard deviation across three seeds.

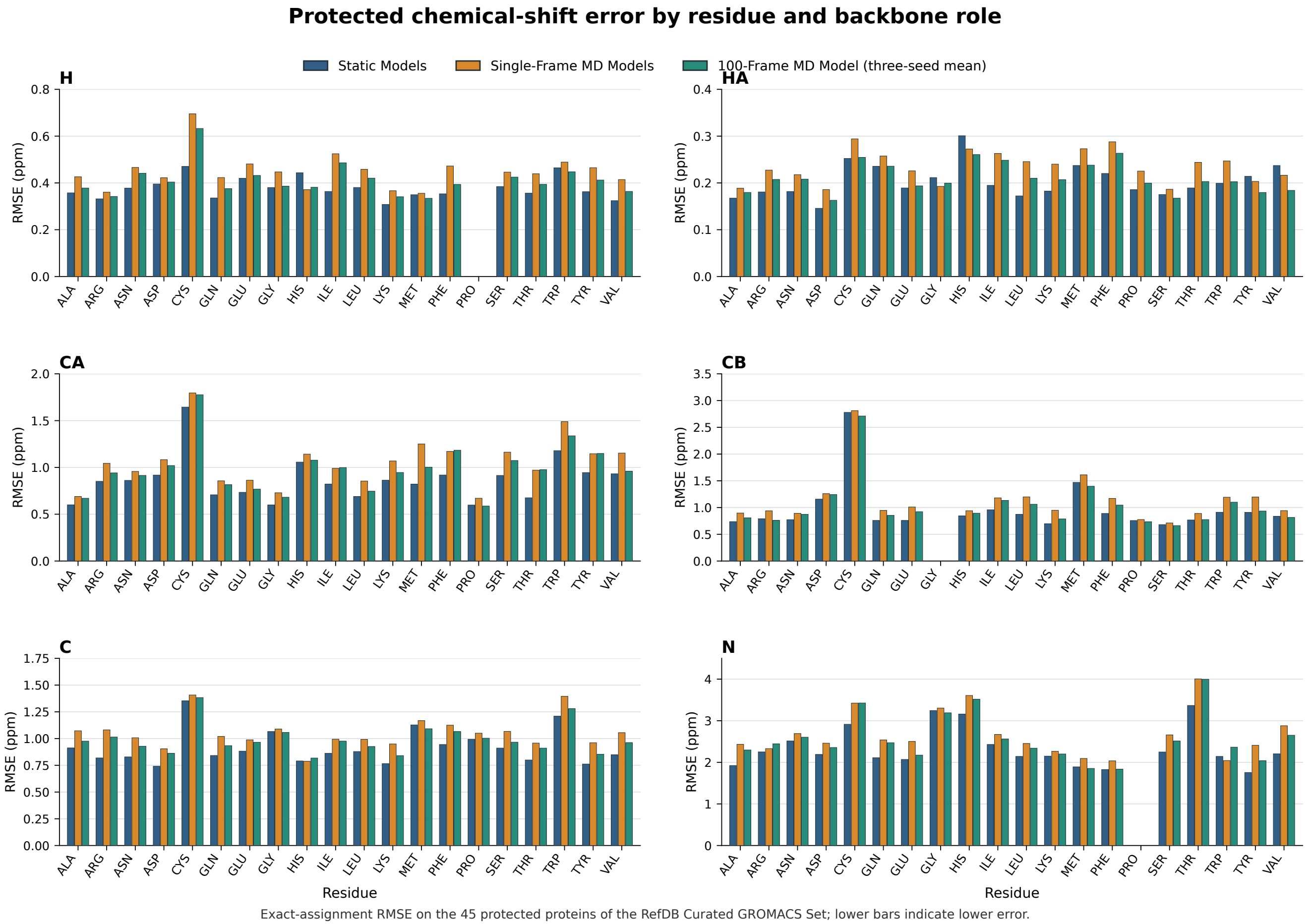


Figure S1: Protected backbone RMSE (ppm) by amino acid for the Static Models, Single-Frame MD Models and three-seed 100-Frame MD Model mean. Whiskers are the sample standard deviation across the three 100-frame fits.

5 External RefDB Results

5.1 RefDB Curated Tleap Set

Model or reference	Atom	Assignments	Proteins	RMSE	MAE
Static Models	H	43,395	381	0.389	0.282
Static Models	HA	41,983	366	0.218	0.151
Static Models	CA	43,434	362	0.873	0.614
Static Models	CB	37,534	355	1.073	0.656
Static Models	C	32,838	298	1.351	0.687
Static Models	N	39,849	367	2.372	1.714
Single-Frame MD Models applied to static structures	H	43,395	381	0.446	0.328
Single-Frame MD Models applied to static structures	HA	41,983	366	0.271	0.184
Single-Frame MD Models applied to static structures	CA	43,434	362	1.027	0.743
Single-Frame MD Models applied to static structures	CB	37,534	355	1.179	0.755
Single-Frame MD Models applied to static structures	C	32,838	298	1.415	0.773
Single-Frame MD Models applied to static structures	N	39,849	367	2.508	1.834
Static residue–atom+DSSP-8 reference	H	43,395	381	0.609	0.460
Static residue–atom+DSSP-8 reference	HA	41,983	366	0.367	0.264
Static residue–atom+DSSP-8 reference	CA	43,434	362	1.406	1.048
Static residue–atom+DSSP-8 reference	CB	37,534	355	1.686	1.081
Static residue–atom+DSSP-8 reference	C	32,838	298	1.638	1.017
Static residue–atom+DSSP-8 reference	N	39,849	367	3.783	2.870
Single-Frame MD residue–atom+DSSP-8 reference applied to static structures	H	43,395	381	0.611	0.461
Single-Frame MD residue–atom+DSSP-8 reference applied to static structures	HA	41,983	366	0.368	0.265
Single-Frame MD residue–atom+DSSP-8 reference applied to static structures	CA	43,434	362	1.410	1.050
Single-Frame MD residue–atom+DSSP-8 reference applied to static structures	CB	37,534	355	1.681	1.080
Single-Frame MD residue–atom+DSSP-8 reference applied to static structures	C	32,838	298	1.644	1.021
Single-Frame MD residue–atom+DSSP-8 reference applied to static structures	N	39,849	367	3.795	2.877

Table S11: Exact-assignment backbone errors (ppm) on the 386-protein low-homology subset of the RefDB Curated Tleap Set.

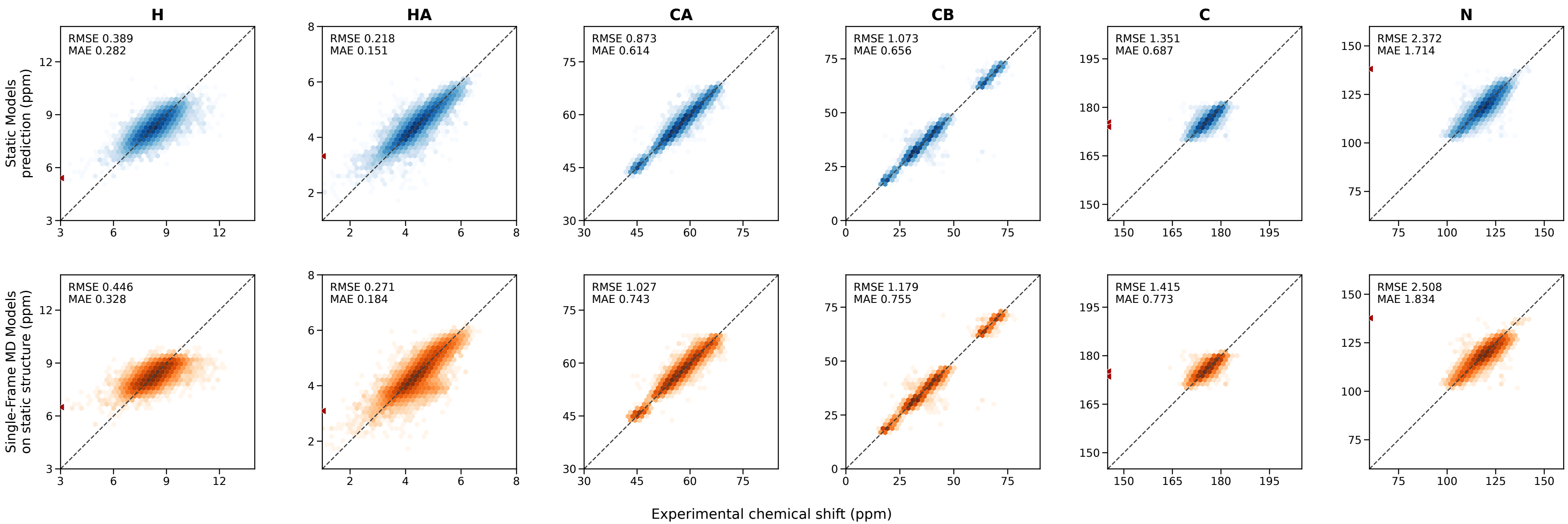
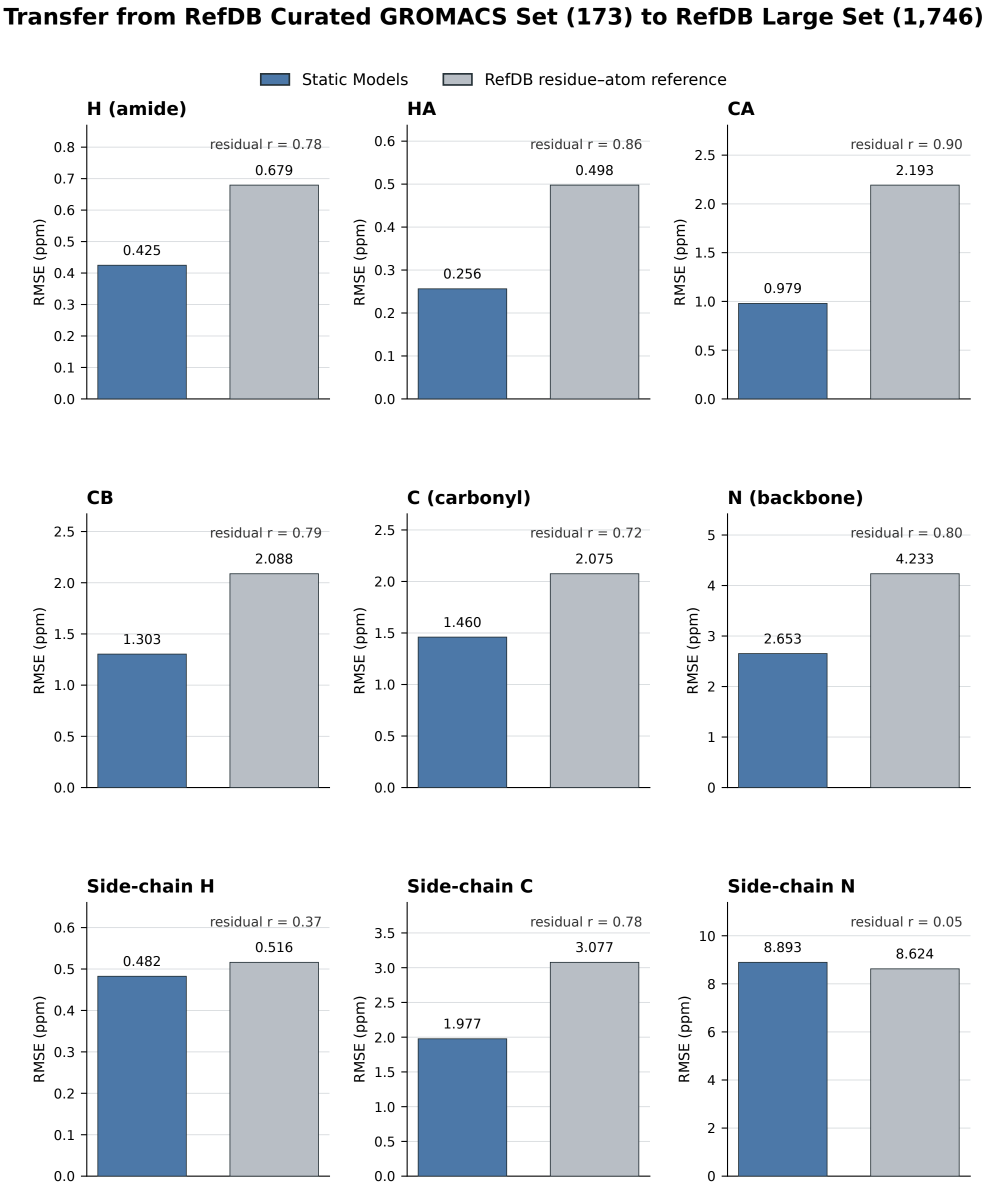


Figure S2: Predicted and experimental backbone shifts for the 386-protein low-homology subset of the RefDB Curated Tleap Set. Dashed lines mark exact agreement; the plotted density and boundary markers follow the frozen reporter.

5.2 RefDB Large Set

Role	Assignments	Proteins	Model RMSE	Model MAE	Reference RMSE	Reference MAE	Δ RMSE	Residual r
H (amide)	179,999	1,719	0.425	0.304	0.679	0.523	37.5%	0.781
HA	140,347	1,460	0.256	0.163	0.498	0.360	48.5%	0.858
CA	149,595	1,244	0.979	0.664	2.193	1.801	55.3%	0.899
CB	122,664	1,176	1.303	0.726	2.088	1.409	37.6%	0.790
C (carbonyl)	106,922	928	1.460	0.783	2.075	1.509	29.6%	0.717
N (backbone)	151,733	1,408	2.653	1.826	4.233	3.187	37.3%	0.798
Side-chain H	203,747	1,440	0.482	0.180	0.516	0.223	6.5%	0.374
Side-chain C	98,931	939	1.977	0.688	3.077	0.871	35.7%	0.780
Side-chain N	9,433	976	8.893	3.048	8.624	3.372	-3.1%	0.046

Table S12: Exact-assignment transfer results for the Static Models on the 1,746-protein low-homology subset of the RefDB Large Set. The reference is the fitting-only RefDB residue–atom mean; Δ RMSE is the percentage reduction from that reference.



Exact-assignment errors; each panel has its own ppm scale. Residual r is Pearson correlation after subtracting the fitting-only RefDB residue–atom reference from prediction and observation.

Figure S3: Static-model and fitting-only RefDB-reference RMSE by reported role on the 1,746-protein low-homology subset of the RefDB Large Set.

Residue	Atom	Assignments	Proteins	Model RMSE	Model MAE	Reference RMSE	Reference MAE	Δ RMSE	Residual r
ALA	H	14,007	1,656	0.383	0.276	0.644	0.495	40.5%	0.804
ALA	HA	10,471	1,400	0.207	0.137	0.474	0.324	56.3%	0.898
ALA	CA	11,291	1,227	0.776	0.533	1.937	1.670	60.0%	0.928
ALA	CB	10,013	1,159	1.162	0.620	1.948	1.311	40.3%	0.809
ALA	C	8,064	910	1.307	0.819	2.105	1.693	37.9%	0.795
ALA	N	12,049	1,386	2.837	1.666	4.015	2.791	29.3%	0.733
ARG	H	8,937	1,620	0.419	0.303	0.668	0.516	37.3%	0.781
ARG	HA	6,885	1,373	0.232	0.159	0.504	0.372	54.1%	0.894
ARG	CA	7,051	1,198	1.005	0.685	2.294	1.975	56.2%	0.910
ARG	CB	6,121	1,128	1.212	0.768	1.901	1.379	36.2%	0.802
ARG	C	5,059	889	1.102	0.779	2.009	1.667	45.2%	0.841
ARG	N	7,423	1,350	2.327	1.663	3.928	3.024	40.8%	0.816
ASN	H	7,836	1,630	0.446	0.322	0.667	0.504	33.2%	0.747
ASN	HA	6,063	1,376	0.220	0.152	0.399	0.297	44.9%	0.837
ASN	CA	6,183	1,193	0.933	0.636	1.789	1.447	47.8%	0.860
ASN	CB	5,522	1,122	0.996	0.692	1.715	1.250	41.9%	0.812
ASN	C	4,356	881	1.122	0.772	1.742	1.406	35.6%	0.776
ASN	N	6,490	1,345	2.600	1.966	4.169	3.196	37.6%	0.815
ASP	H	11,068	1,647	0.437	0.313	0.625	0.469	30.1%	0.714
ASP	HA	8,361	1,388	0.196	0.141	0.336	0.244	41.6%	0.825
ASP	CA	8,892	1,210	0.887	0.604	1.992	1.677	55.5%	0.900
ASP	CB	7,931	1,144	1.500	0.682	1.898	1.133	20.9%	0.620
ASP	C	6,354	899	1.091	0.742	1.693	1.352	35.6%	0.774
ASP	N	9,553	1,370	2.517	1.860	4.070	3.123	38.2%	0.820
CYS	H	4,332	1,214	0.518	0.369	0.712	0.555	27.2%	0.690
CYS	HA	3,774	1,033	0.560	0.219	0.761	0.463	26.4%	0.671
CYS	CA	2,412	845	1.659	1.180	3.412	2.651	51.4%	0.870
CYS	CB	2,115	782	3.552	2.029	7.043	6.465	49.6%	0.837
CYS	C	1,757	632	1.418	0.984	2.003	1.634	29.2%	0.742
CYS	N	2,832	957	2.924	2.216	4.521	3.580	35.3%	0.796
GLN	H	7,491	1,564	0.386	0.285	0.627	0.488	38.4%	0.794
GLN	HA	5,879	1,315	0.208	0.140	0.459	0.335	54.7%	0.893
GLN	CA	6,045	1,181	0.893	0.621	2.129	1.840	58.1%	0.915
GLN	CB	5,396	1,115	1.063	0.668	1.840	1.367	42.3%	0.819
GLN	C	4,299	878	0.953	0.686	1.935	1.623	50.8%	0.871
GLN	N	6,356	1,321	2.189	1.628	3.768	2.889	41.9%	0.833
GLU	H	14,014	1,654	0.385	0.280	0.636	0.489	39.5%	0.797
GLU	HA	10,649	1,398	0.193	0.131	0.449	0.320	57.1%	0.904
GLU	CA	11,489	1,236	0.913	0.630	2.088	1.779	56.3%	0.907
GLU	CB	10,150	1,163	1.053	0.630	1.774	1.253	40.6%	0.813
GLU	C	8,429	920	2.052	0.731	2.583	1.610	20.5%	0.610
GLU	N	12,179	1,397	2.473	1.702	3.830	2.827	35.4%	0.801
GLY	H	13,701	1,679	0.443	0.308	0.682	0.496	35.0%	0.762
GLY	HA	6,916	887	0.430	0.258	0.417	0.260	-2.9%	0.364
GLY	CA	10,685	1,229	0.727	0.505	1.165	0.885	37.5%	0.792
GLY	C	7,570	913	2.639	0.896	2.897	1.405	8.9%	0.423
GLY	N	11,299	1,389	3.128	1.834	4.194	3.028	25.4%	0.682
HIS	H	4,061	1,411	0.513	0.363	0.726	0.564	29.3%	0.715
HIS	HA	3,139	1,175	0.297	0.189	0.510	0.362	41.7%	0.812

continued

Residue	Atom	Assignments	Proteins	Model RMSE	Model MAE	Reference RMSE	Reference MAE	Δ RMSE	Residual r
HIS	CA	3,301	1,077	1.290	0.909	2.350	1.916	45.1%	0.839
HIS	CB	2,926	1,002	1.559	1.083	2.268	1.647	31.3%	0.714
HIS	C	2,332	789	1.207	0.857	1.985	1.616	39.2%	0.799
HIS	N	3,315	1,168	2.683	2.010	4.272	3.306	37.2%	0.807
ILE	H	9,937	1,628	0.413	0.303	0.721	0.572	42.7%	0.821
ILE	HA	7,514	1,374	0.222	0.165	0.591	0.476	62.5%	0.932
ILE	CA	8,121	1,204	1.092	0.785	2.617	2.215	58.3%	0.911
ILE	CB	7,194	1,142	1.056	0.717	1.982	1.521	46.7%	0.848
ILE	C	5,817	896	1.018	0.724	1.829	1.488	44.3%	0.832
ILE	N	8,575	1,350	2.556	1.863	4.520	3.436	43.5%	0.829
LEU	H	16,096	1,681	0.418	0.295	0.704	0.549	40.7%	0.806
LEU	HA	12,081	1,418	0.193	0.135	0.503	0.389	61.6%	0.925
LEU	CA	13,137	1,228	0.841	0.575	2.061	1.808	59.2%	0.922
LEU	CB	11,629	1,155	1.195	0.865	1.860	1.407	35.7%	0.836
LEU	C	9,551	914	1.113	0.771	1.920	1.582	42.0%	0.825
LEU	N	13,765	1,391	2.296	1.690	4.108	3.219	44.1%	0.850
LYS	H	13,706	1,659	0.386	0.281	0.648	0.500	40.4%	0.804
LYS	HA	10,438	1,401	0.217	0.142	0.469	0.342	53.8%	0.887
LYS	CA	10,890	1,208	0.901	0.621	2.160	1.859	58.3%	0.916
LYS	CB	9,574	1,137	1.166	0.668	1.858	1.264	37.2%	0.792
LYS	C	7,848	898	1.565	0.743	2.235	1.593	30.0%	0.715
LYS	N	11,590	1,369	2.429	1.788	3.916	3.023	38.0%	0.826
MET	H	3,644	1,340	0.428	0.296	0.663	0.508	35.4%	0.765
MET	HA	2,838	1,120	0.236	0.167	0.522	0.384	54.7%	0.891
MET	CA	3,176	1,054	1.100	0.785	2.188	1.824	49.7%	0.886
MET	CB	2,775	981	1.503	0.981	2.339	1.715	35.7%	0.775
MET	C	2,194	758	1.200	0.813	2.028	1.676	40.8%	0.815
MET	N	3,187	1,152	2.749	1.779	4.116	2.980	33.2%	0.752
PHE	H	7,030	1,568	0.462	0.339	0.753	0.595	38.7%	0.791
PHE	HA	5,319	1,306	0.257	0.183	0.588	0.462	56.3%	0.900
PHE	CA	5,726	1,180	1.155	0.816	2.581	2.206	55.3%	0.893
PHE	CB	5,133	1,108	1.079	0.755	2.027	1.579	46.8%	0.855
PHE	C	4,057	877	1.200	0.842	1.951	1.571	38.5%	0.808
PHE	N	6,031	1,329	2.434	1.815	4.318	3.352	43.6%	0.846
PRO	HA	5,998	1,311	0.234	0.154	0.377	0.242	37.9%	0.797
PRO	CA	6,210	1,172	0.830	0.550	1.446	1.125	42.6%	0.825
PRO	CB	5,493	1,100	1.010	0.522	1.215	0.695	16.9%	0.565
PRO	C	4,289	856	1.074	0.779	1.518	1.163	29.2%	0.705
PRO	N	125	39	16.379	7.541	16.126	7.835	-1.6%	-0.133
SER	H	11,698	1,681	0.419	0.296	0.631	0.475	33.6%	0.750
SER	HA	9,312	1,422	0.228	0.153	0.433	0.307	47.4%	0.855
SER	CA	9,330	1,223	0.989	0.678	2.047	1.683	51.7%	0.882
SER	CB	8,108	1,156	0.999	0.585	1.507	1.071	33.7%	0.749
SER	C	6,734	914	1.168	0.768	1.704	1.306	31.4%	0.749
SER	N	9,533	1,383	2.468	1.835	3.803	2.932	35.1%	0.793
THR	H	10,575	1,642	0.436	0.317	0.673	0.521	35.2%	0.762
THR	HA	8,152	1,388	0.232	0.164	0.509	0.392	54.3%	0.890
THR	CA	8,284	1,202	1.110	0.732	2.571	2.051	56.8%	0.903
THR	CB	7,261	1,137	2.057	0.731	2.368	1.262	13.1%	0.502
THR	C	5,782	889	1.888	0.794	2.258	1.372	16.4%	0.554

continued

Residue	**Atom**	**Assignments**	**Proteins**	**Model RMSE**	**Model MAE**	**Reference RMSE**	**Reference MAE**	**Δ RMSE**	**Residual *r***
THR	N	8,868	1,356	3.104	2.243	5.080	3.984	38.9%	0.818
TRP	H	2,495	1,137	0.521	0.379	0.826	0.657	36.9%	0.772
TRP	HA	1,893	923	0.280	0.203	0.561	0.435	50.2%	0.865
TRP	CA	1,914	851	1.311	0.951	2.525	2.084	48.1%	0.868
TRP	CB	1,702	786	1.192	0.837	1.936	1.466	38.4%	0.808
TRP	C	1,408	633	1.179	0.859	1.957	1.573	39.8%	0.821
TRP	N	2,049	923	3.294	1.917	4.944	3.442	33.4%	0.764
TYR	H	6,139	1,593	0.448	0.327	0.767	0.614	41.7%	0.814
TYR	HA	4,665	1,339	0.267	0.182	0.599	0.462	55.3%	0.896
TYR	CA	4,739	1,160	1.141	0.806	2.451	2.047	53.4%	0.885
TYR	CB	4,148	1,083	1.164	0.786	2.134	1.688	45.4%	0.853
TYR	C	3,360	864	1.122	0.807	1.921	1.545	41.6%	0.818
TYR	N	5,052	1,313	2.447	1.807	4.454	3.474	45.1%	0.844
VAL	H	13,232	1,655	0.423	0.299	0.721	0.569	41.3%	0.811
VAL	HA	10,000	1,396	0.291	0.215	0.600	0.485	51.5%	0.880
VAL	CA	10,719	1,222	1.083	0.708	2.787	2.353	61.1%	0.923
VAL	CB	9,473	1,157	0.971	0.599	1.758	1.414	44.8%	0.834
VAL	C	7,662	915	1.074	0.757	1.831	1.471	41.4%	0.816
VAL	N	11,462	1,377	2.576	1.866	4.691	3.608	45.1%	0.841

Table S13: Residue–backbone exact-assignment transfer results on the 1,746-protein low-homology subset of the RefDB Large Set.